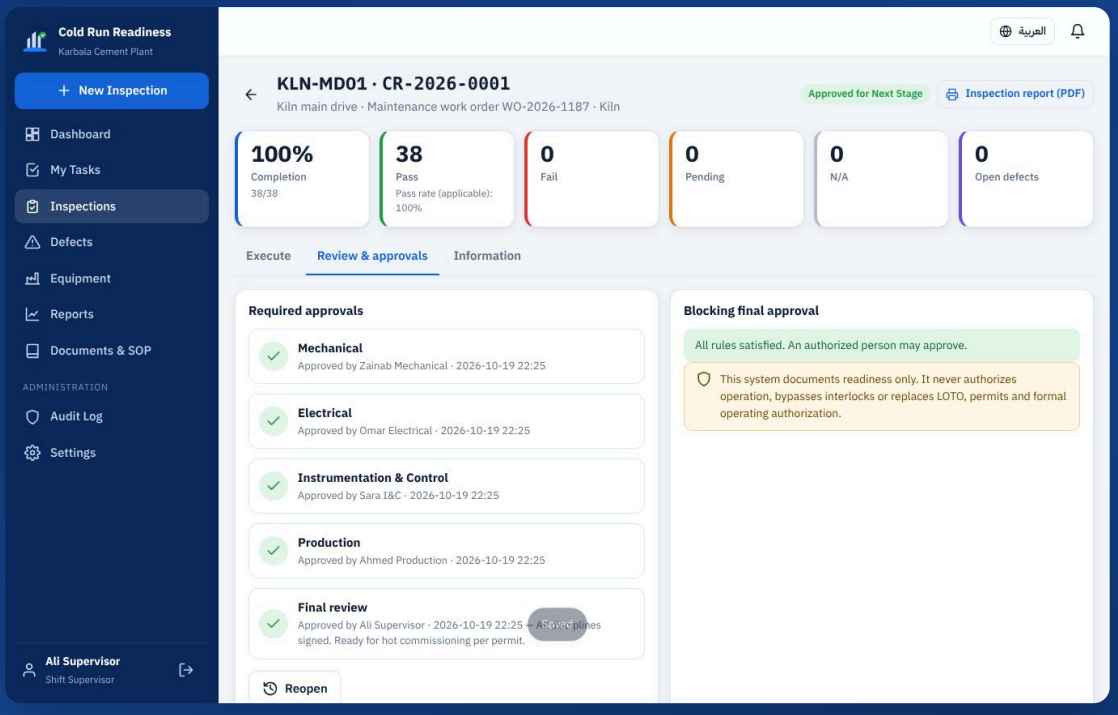
Karbala Cement Plant · Delta Company

Cold Run Inspection & Readiness Management Platform

منصة إدارة فحص وجاهزية التشغيل البارد (Cold Run) بعد الصيانة



Project documentation — what was built, how it works, screenshots and deployment guide

Version 1.0 · 19 October 2026 · Repository `yosx1/new` → folder `platform/`

Contents

1. Executive summary (English / العربية)
2. What was delivered
3. Architecture and Firebase free plan
4. User roles and task pools
5. Workflows and business rules
6. Screen-by-screen walkthrough (with screenshots)
7. Arabic, mobile and offline use
8. Security, approvals and audit trail
9. Testing performed and results
10. Deployment to Firebase Hosting
11. Limitations and recommended next steps

1. Executive summary

The platform replaces paper Cold Run checklists with a central, role-based system that records equipment readiness after maintenance. Shift supervisors create an inspection from a maintenance work order; mechanical, electrical, instrumentation & control and production engineers complete their own checklist sections from a phone, tablet or computer; failed checks become tracked defects with corrective actions; each discipline signs off; and an authorized supervisor gives the final "approved for next stage" decision. Management sees plant-wide readiness, defects and approvals on a live dashboard and in ten reports.

It runs in Arabic and English, installs on phones as an app, keeps working without network, and is designed to run on the **Firebase free (Spark) plan**. All permissions are enforced on the server by Firestore security rules and are covered by automated tests.

الملخص التنفيذي

تستبدل المنصة قوائم فحص الـ Cold Run الورقية بنظام مركزي يعتمد على الأدوار لتوثيق جاهزية المعدات بعد الصيانة. ينشئ مسؤول الوجبة الفحص من أمر عمل الصيانة، ويُكمل مهندسو الميكانيك والكهرباء والسيطرة والإنتاج أقسامهم من الهاتف أو الجهاز اللوحي أو الحاسوب، وتتحول البنود المرفوضة إلى عيوب بإجراءات تصحيحية متابَعة، ثم يوقّع كل اختصاص، ويصدر الشخص المخوّل قرار الاعتماد للمرحلة التالية. وتتابع الإدارة جاهزية المعمل والعيوب والموافقات عبر لوحة متابعة مباشرة وعشرة تقارير. يعمل النظام باللغتين العربية والإنكليزية، ويُثبّت على الهاتف كتطبيق، ويستمر بالعمل دون اتصال، ومصمم للعمل على خطة Firebase المجانية، مع فرض جميع الصلاحيات على الخادم.

Safety statement. The system documents readiness only. It never authorizes operation, bypasses interlocks, or replaces LOTO, permits and formal operating authorization. The example checklist items are not approved engineering acceptance criteria and must be confirmed against plant procedures and equipment manuals.

2. What was delivered

Module	Delivered functionality
Dashboard	Active inspections, awaiting execution, in progress, awaiting corrective action, awaiting approval, approved, critical open defects, overdue corrective actions, "waiting in your pool", readiness by department, status chart, recent defects, follow-ups and activity. Filters: department, equipment, dates, status, discipline.
My Tasks (pools)	Checklist sections, defects, verifications, departmental approvals and final reviews routed to the user's role. "Take it" / "Return to pool" lets any colleague on shift continue the work.
New inspection wizard	3 steps (information → checklist → review). Automatic ID `CR-YYYY-NNNN`, all required fields with validation, draft auto-save, optional work-order attachments, automatic checklist selection by equipment and maintenance type.
Inspection execution	Sections grouped by stage and discipline; one-tap Pass / Fail / N/A; detail sheet with measured value, unit and equipment-specific limits, N/A reason, before/after photos, files, assignee, supervisor comment; progress saved per item.
Defects & corrective actions	Unique `DF-YYYY-NNNN`, severity, responsible discipline, assignee, target date, corrective action, photos; 6-stage workflow with independent verification and reopen; critical, overdue, per-discipline views; recurring-failure flag.
Approvals	Required disciplines per checklist template plus final authorized review; automatic blockers; documented override restricted to configured roles.
History & search	Search by ID, work order, tag, equipment, department, date, inspector, defect ID, status, maintenance type; equipment history with previous inspections, repeated defects, measurements, approvals and photos.
Reports	10 reports with CSV export and print/PDF, plus an individual A4 inspection report with signatures block, document version and generation time.
Administration	Users & roles, devices (revoke = immediate sign-out), departments & equipment hierarchy, versioned checklist templates, document/SOP library, system settings, audit log.
Deployment	Firebase Hosting configuration, Firestore rules & indexes, GitHub Actions for CI (type-check, build, rules tests) and automatic deployment to Firebase on merge to `main`.

Earlier in the project a single-file prototype (`index.html`, based on the original mockup) was also delivered; the new platform in `platform/` supersedes it.

3. Architecture and Firebase free plan

Layer	Technology
Front-end	React 18 + TypeScript, Vite build, route-level code splitting, installable PWA with service worker (app shell cached for offline start).
Authentication	Firebase Authentication (email/password). First visitor creates the System Administrator; admins create other accounts.
Database	Cloud Firestore with persistent local cache (IndexedDB) — data is cached on each device and only changes are downloaded.
Business rules & security	`firestore.rules` — roles, workflow transitions, approvals, defect closure, append-only audit, device revocation.
Hosting	Firebase Hosting (`https://<project>.web.app`), SPA rewrites and long-term caching of hashed assets.

Free-plan (Spark) design decisions

- Cloud Functions and (for new projects) Cloud Storage require the paid Blaze plan, so **no server code is used**: all rules run in Firestore Security Rules, and new user accounts are created from the admin's browser through a secondary Firebase app.
- Photos are resized and compressed on the phone (≈100–250 KB) and stored as Firestore documents; files up to ≈650 KB can be attached; large manuals are linked by URL.
- Quota-aware: the newest 400 inspections and 800 defects stay live, older history is loaded on demand, device heartbeats are throttled, and the local cache avoids repeated downloads.

Data model (Firestore collections)

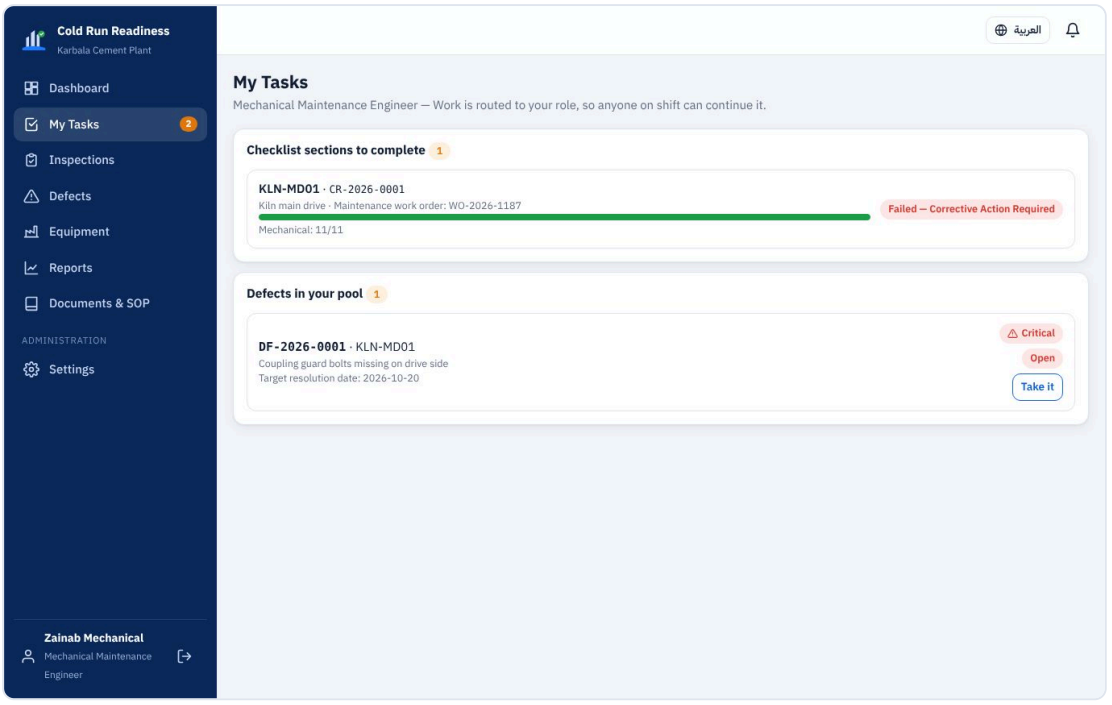
Collection	Purpose
`users` , `devices`	Profiles, roles, department scope, active flag; every signed-in device with last-seen and revoked flag.
`departments` , `equipment`	Plant → department → line → equipment → sub-equipment (`parentId`), linked checklist templates.
`templates/{id}/versions/{n}`	Checklist templates; each published version is immutable.
`inspections` + `results/{item}`	Inspection header, checklist snapshot, summary, status, approvals, task pools; one result document per checklist item.
`defects` , `attachments` , `documents`	Defects & corrective actions; photos/files; SOP and document library.
`notifications` , `audit` , `counters` , `config`	In-app notifications; append-only audit trail; sequential IDs; settings.

4. User roles and task pools

Role	Main permissions
System Administrator	Users, roles and devices; departments and equipment; checklist templates and versions; settings and override roles; all reports and audit log.
Shift Supervisor	Create and assign inspections, monitor progress, assign and verify defects, final readiness approval within scope.
Department Manager	Departmental approvals, final approval, documented override (by default), verification, readiness monitoring.
Mechanical / Electrical / I&C Engineer	Complete own discipline's checklist sections, record measurements and evidence, work on defects in own pool, departmental approval for own discipline.
Production Engineer / Operator	Prerequisites and Cold Run operational test sections, readings, production approval.
Viewer	Read-only access to dashboards, records and reports.

Task pools — work follows the role, not the person

Every inspection stores which pools currently need it (e.g. `exec:mechanical`, `approve:electrical`, `review`). Every defect belongs to the pool of its responsible discipline. When a person changes shift, goes on leave, or is deactivated, colleagues with the same role see the same work in **My Tasks** and can take it over. The administrator's **Reassign open work** action (and deactivation) returns a person's assigned defects to the pool.



My Tasks for a second mechanical engineer: the defect raised by a colleague appears in the pool with a *Take it* button.

5. Workflows and business rules

Inspection status

Draft → Maintenance Completed → Inspection in Progress → Failed — Corrective Action Required → Awaiting Verification → Awaiting Departmental Approval → Ready for Authorized Review → Approved for Next Stage · Rejected / Reopened

The status is derived automatically from recorded results, defects and approvals; only "approved" and "rejected" are explicit human decisions.

Defect & corrective action

Open → Assigned → In Progress → Repair Completed → Awaiting Verification → Verified & Closed

- Only supervisors, managers or administrators can close a defect, only with a **passing verification**, and **never the person who recorded the repair**. A failed verification sends it back to *In Progress*; closed defects can be reopened by authorized roles.
- After a defect is closed, the failed checklist item must be re-inspected.

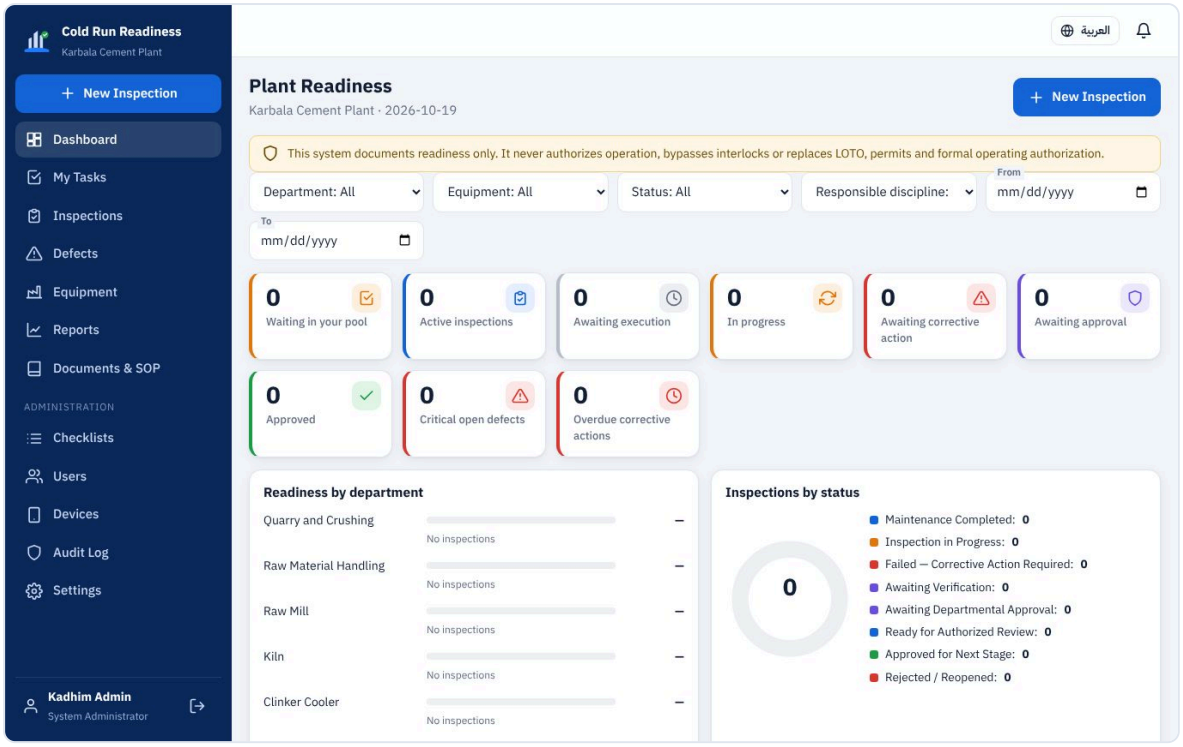
Readiness and approval rules

- Pass rate excludes N/A items from the denominator.
- Final approval is blocked by: pending or mandatory items, failed items not re-inspected, open defects, open critical defects, critical failures, and missing departmental approvals.
- Critical blockers can only be overridden by the roles set in Settings (default: Manager, Administrator), with a written justification stored on the record and in the audit log.
- Department readiness = share of inspected equipment whose latest inspection is approved; equipment with an open critical defect is never counted as ready.
- Required departmental approvals are configured per checklist template (e.g. pumps need mechanical + electrical only).

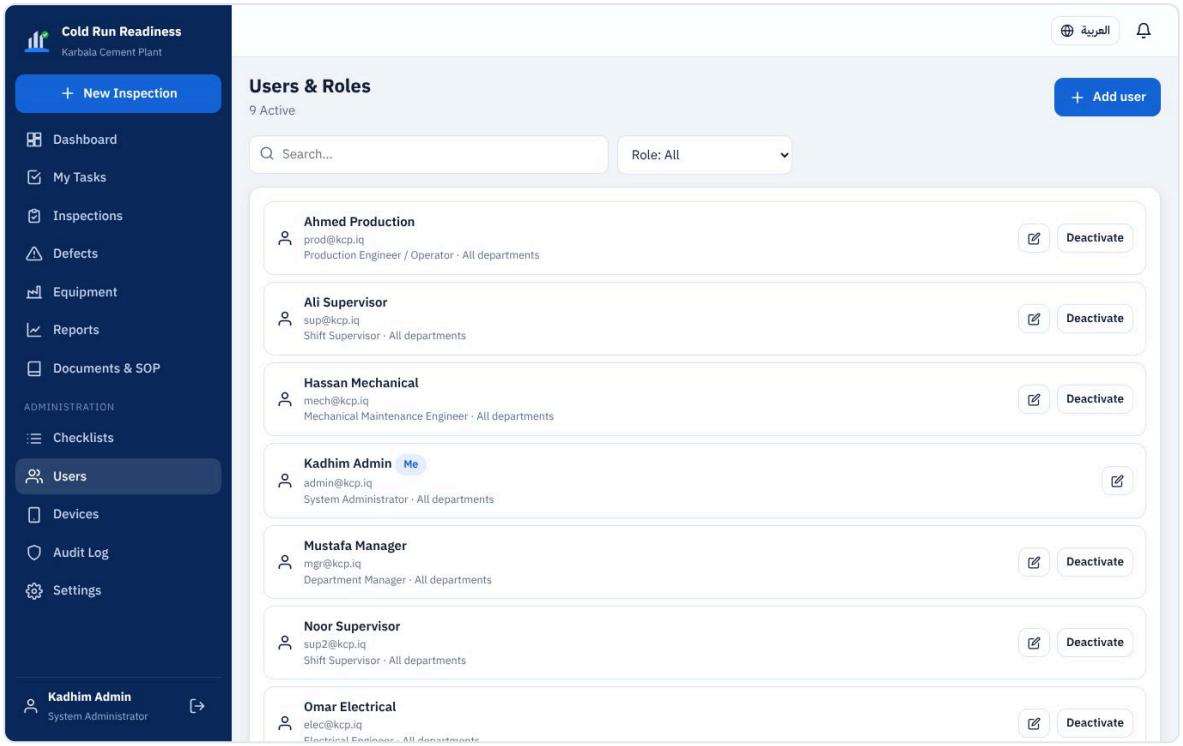
6. Screen-by-screen walkthrough

6.1 First-time setup and sign-in

When no administrator exists, the first visitor creates the System Administrator account. The administrator then loads the example departments, equipment and checklists from Settings (to be reviewed and replaced by plant engineering) and creates users.

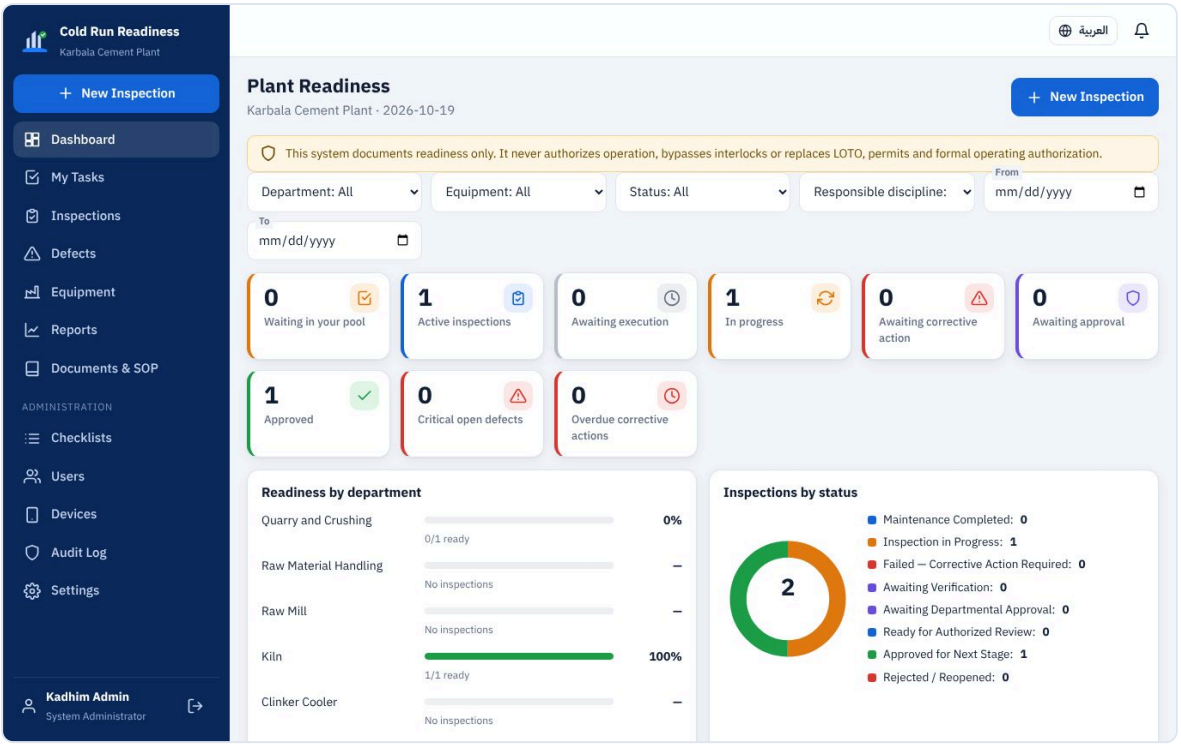


Dashboard right after setup — 8 departments loaded, no inspections yet. The safety statement is always visible.



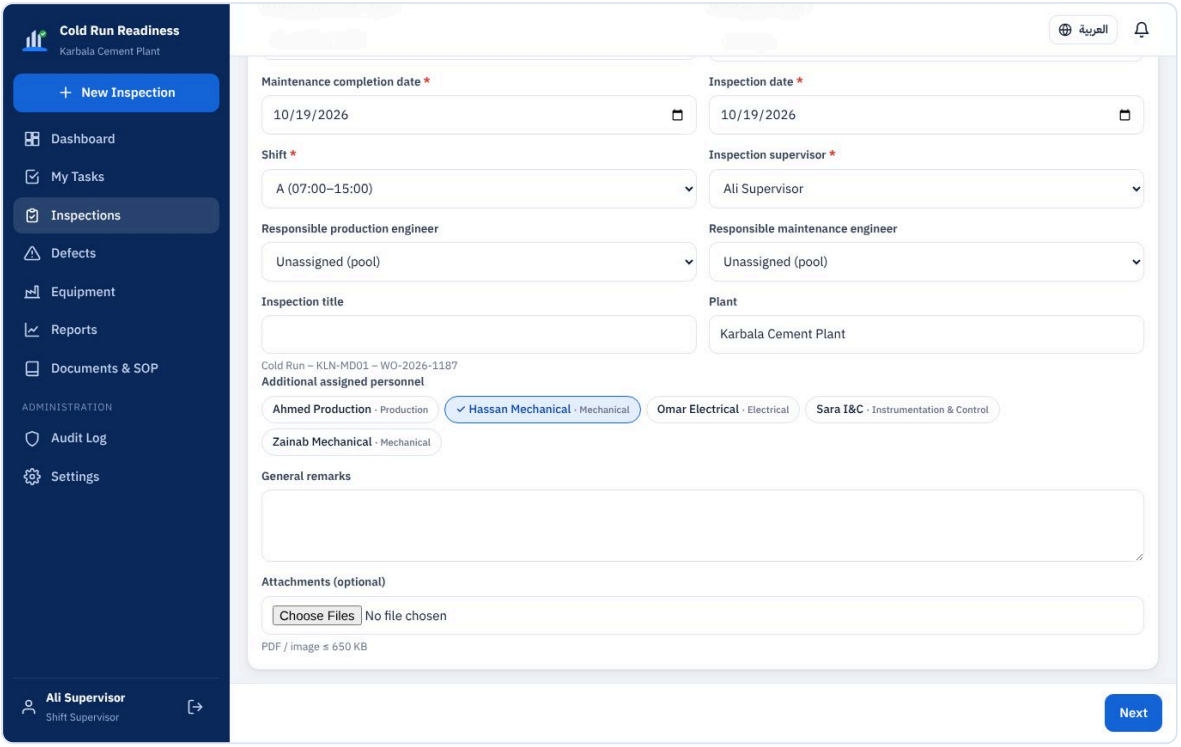
Users & Roles — create accounts, set role and department scope, send password reset, deactivate, and reassign open work.

6.2 Dashboard (management view)

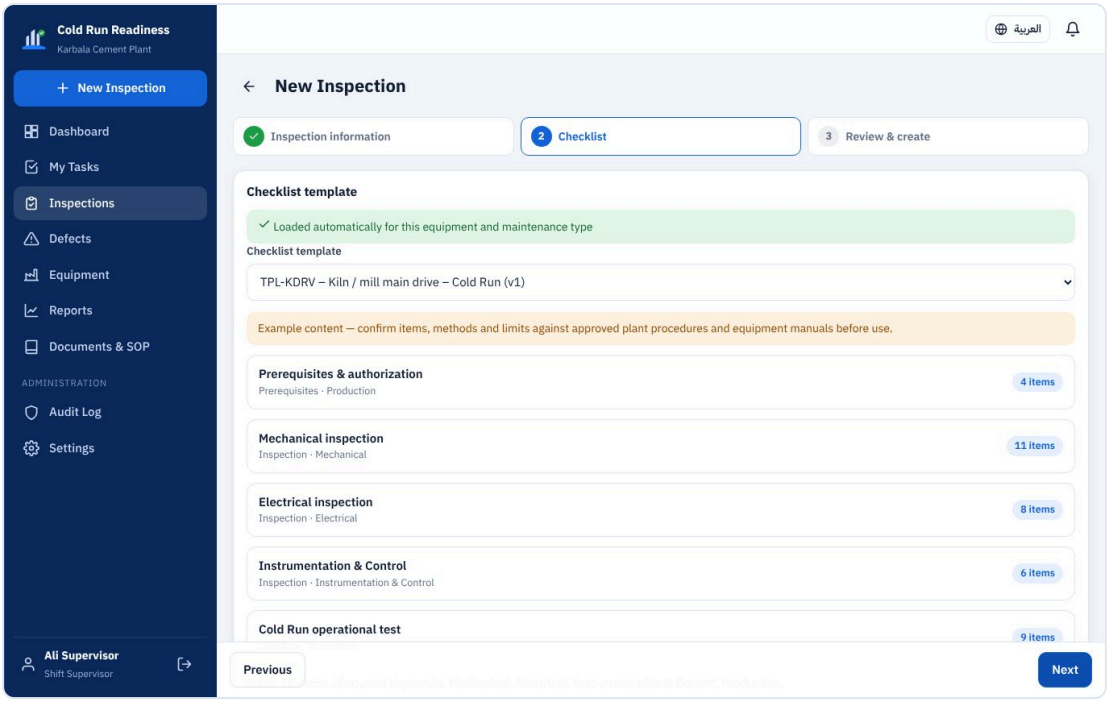


Plant Readiness dashboard — KPI cards (clickable), filters, readiness by department and inspections by status. Kiln shows 50%: the main drive is approved, the rotary kiln is not yet.

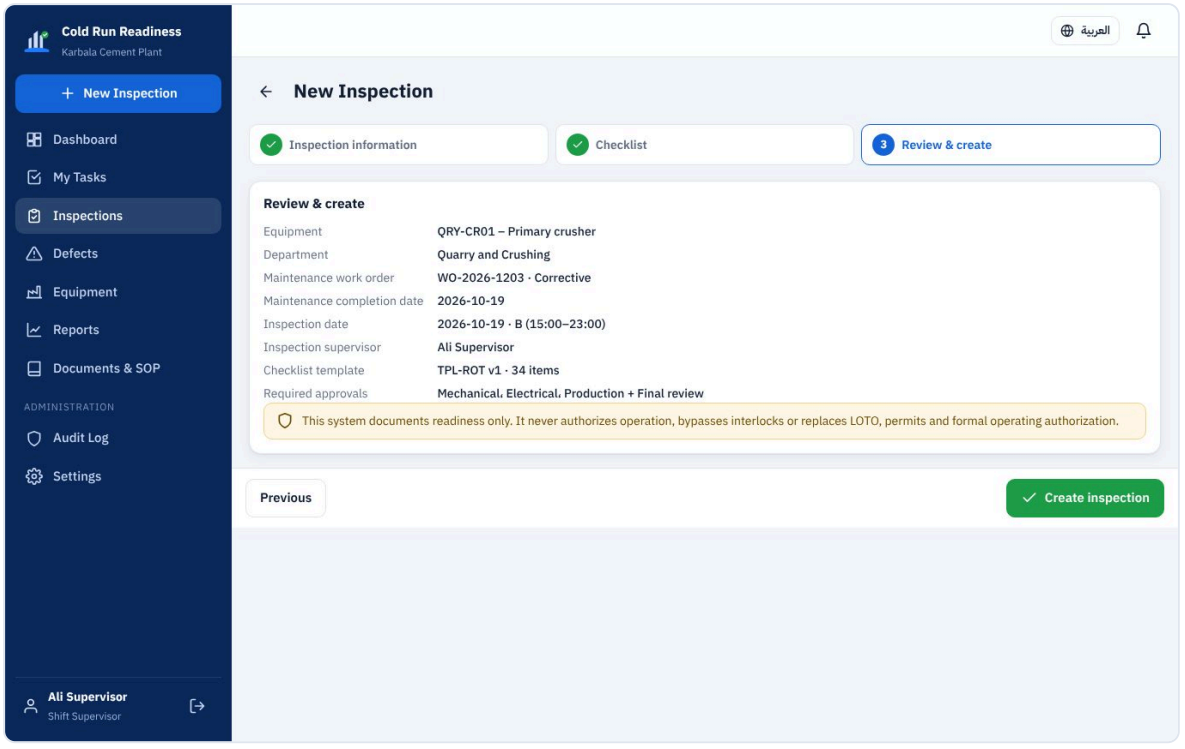
6.3 Creating an inspection



Step 1 – Inspection information: department, equipment, work order, maintenance type, dates, shift, supervisor, responsible engineers, additional personnel, remarks, attachments.

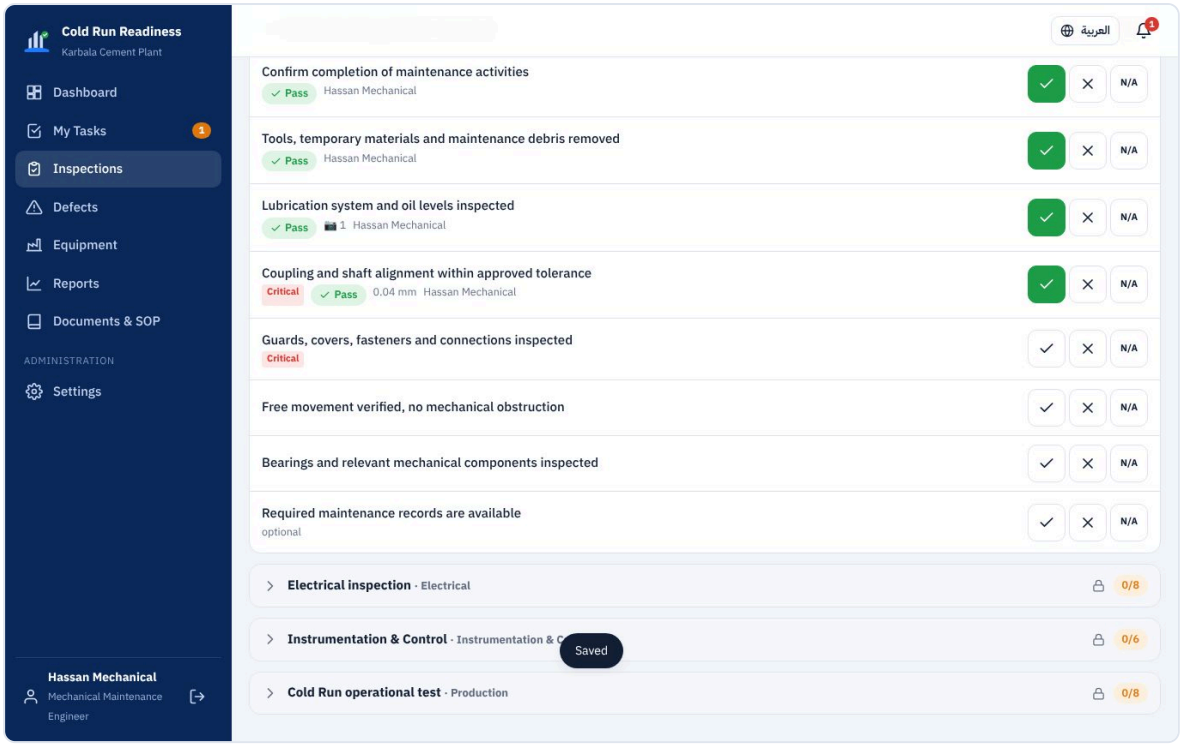


Step 2 – Checklist loaded automatically for the Kiln main drive (5 sections, item counts and required approvals).

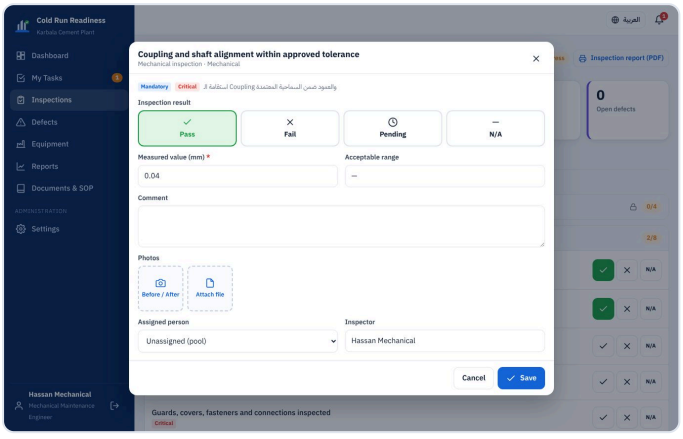


Step 3 – Review & create with the safety statement. The ID (CR-2026-NNNN) is generated on creation.

6.4 Executing the checklist



Execution view — KPIs, sections per discipline (sections of other disciplines are read-only with a lock icon), one-tap Pass / Fail / N/A buttons, critical and optional flags, inspector name.



Measurement item — value with unit and equipment-specific acceptable range.

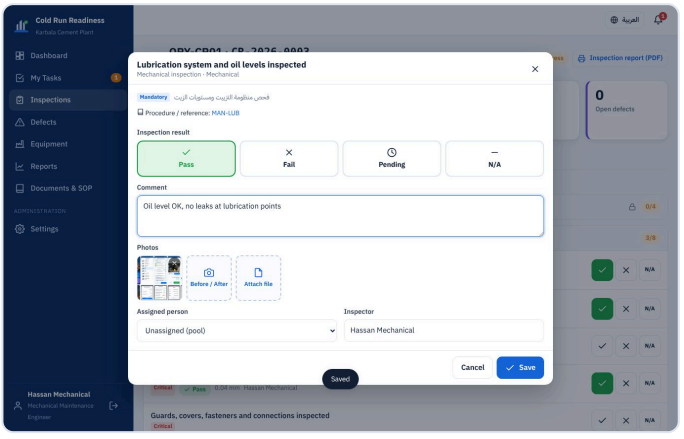
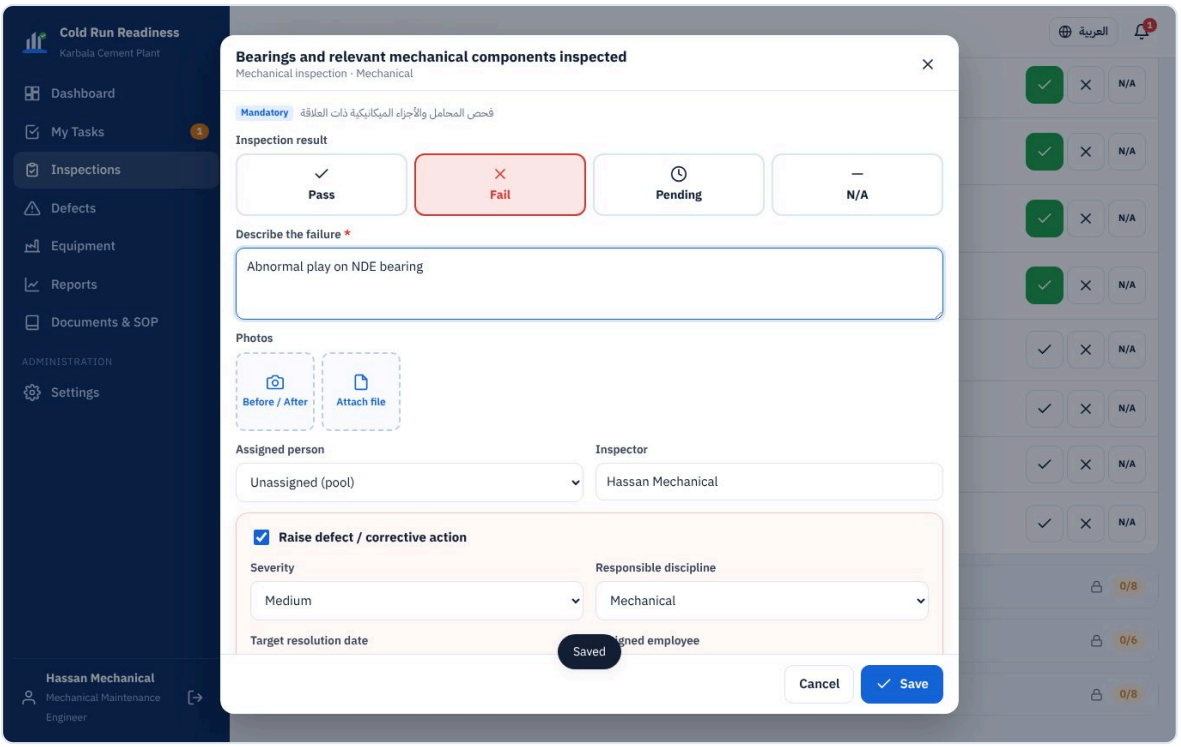
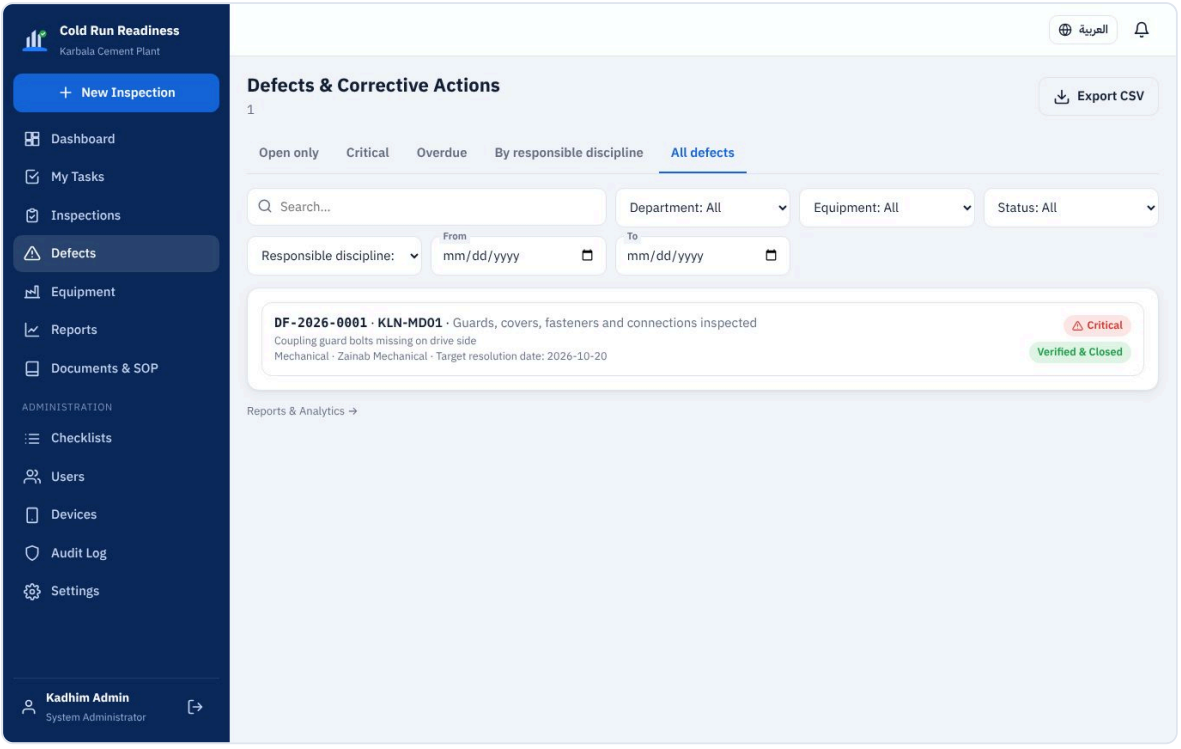


Photo evidence — camera capture, compressed on the device.

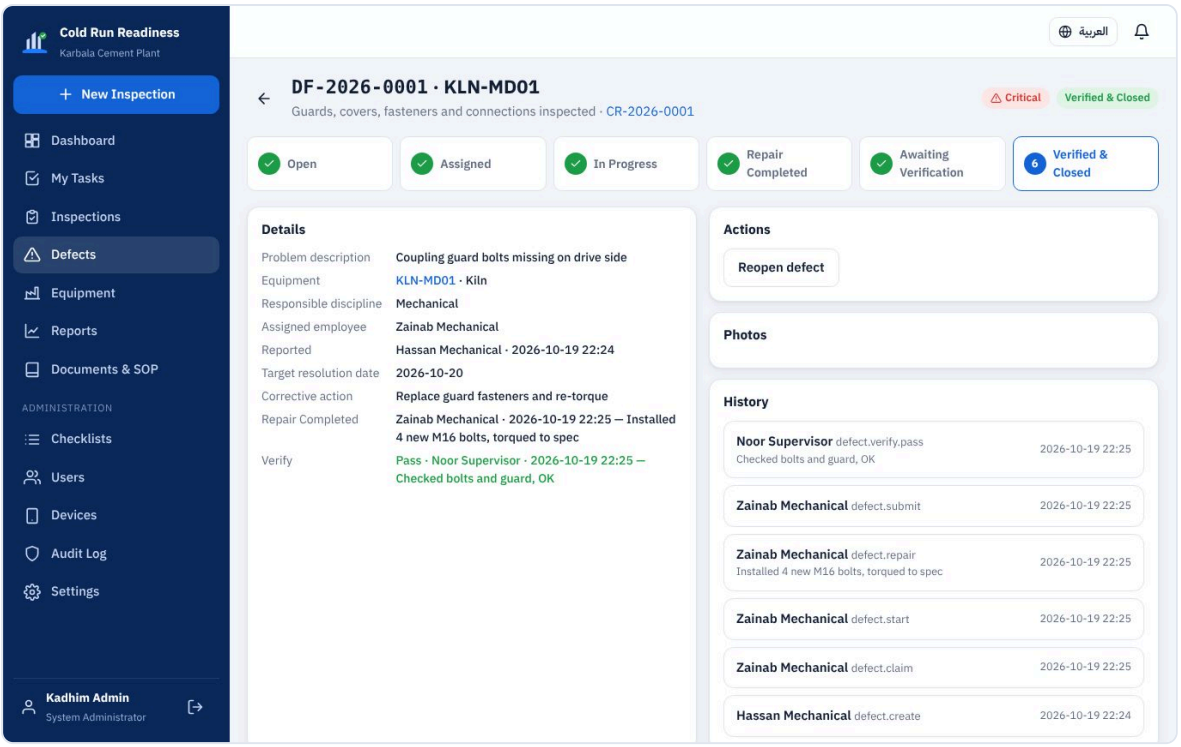


Fail requires a description and opens the defect form: severity, responsible discipline, target date (default from severity), assignee and corrective action.

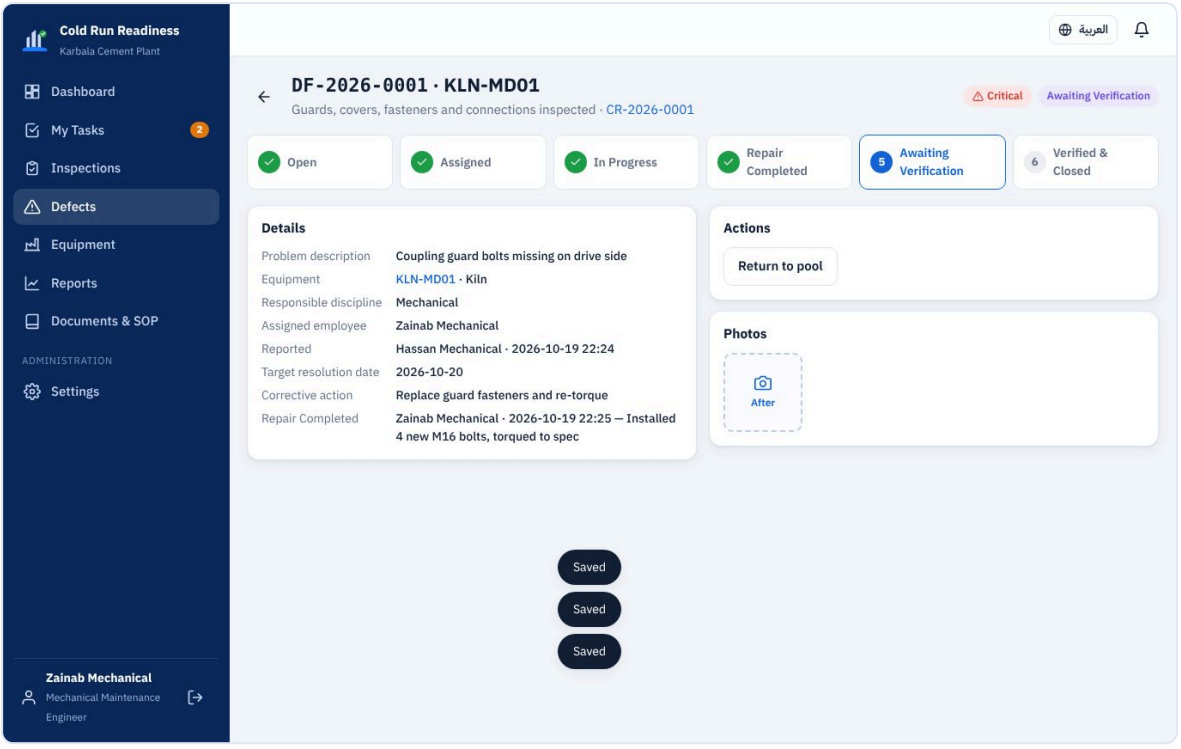
6.5 Defects and corrective actions



Defects — tabs for open, critical, overdue, by responsible discipline and all; filters; CSV export; recurring-failure badge.

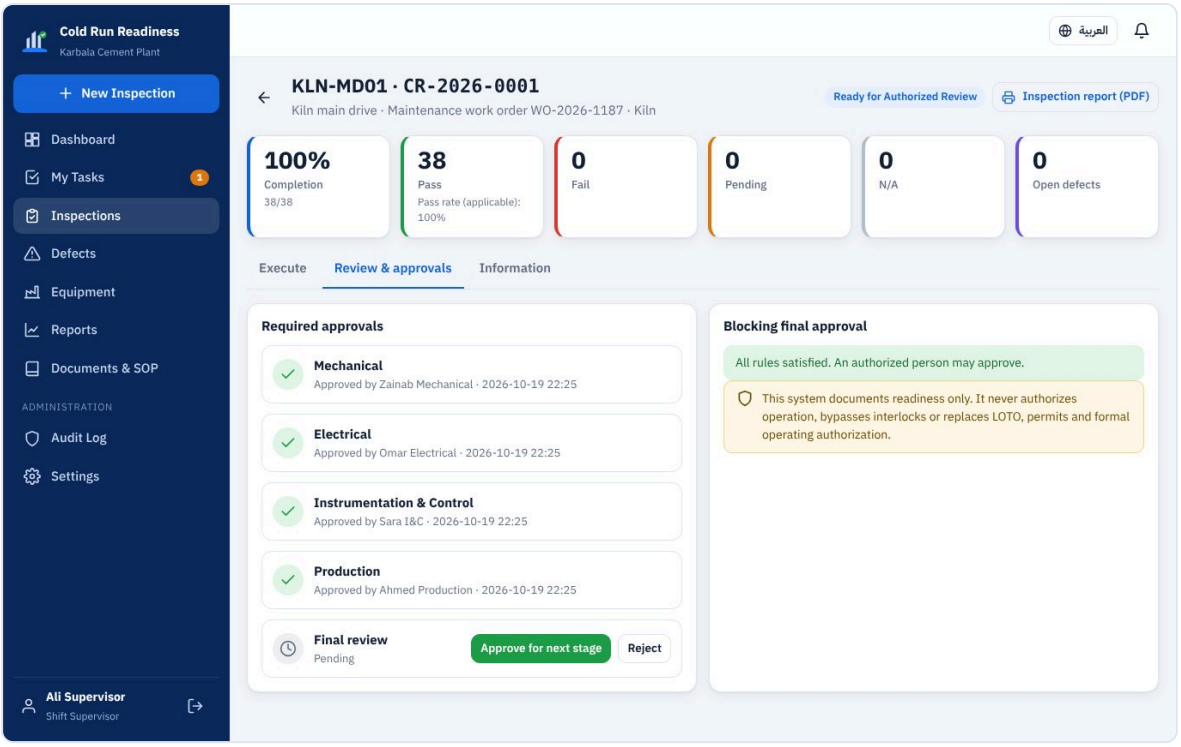


Defect detail — workflow stepper, details, repair and verification records, photos and full history. Here: repaired by Zainab, verified and closed by a different supervisor.

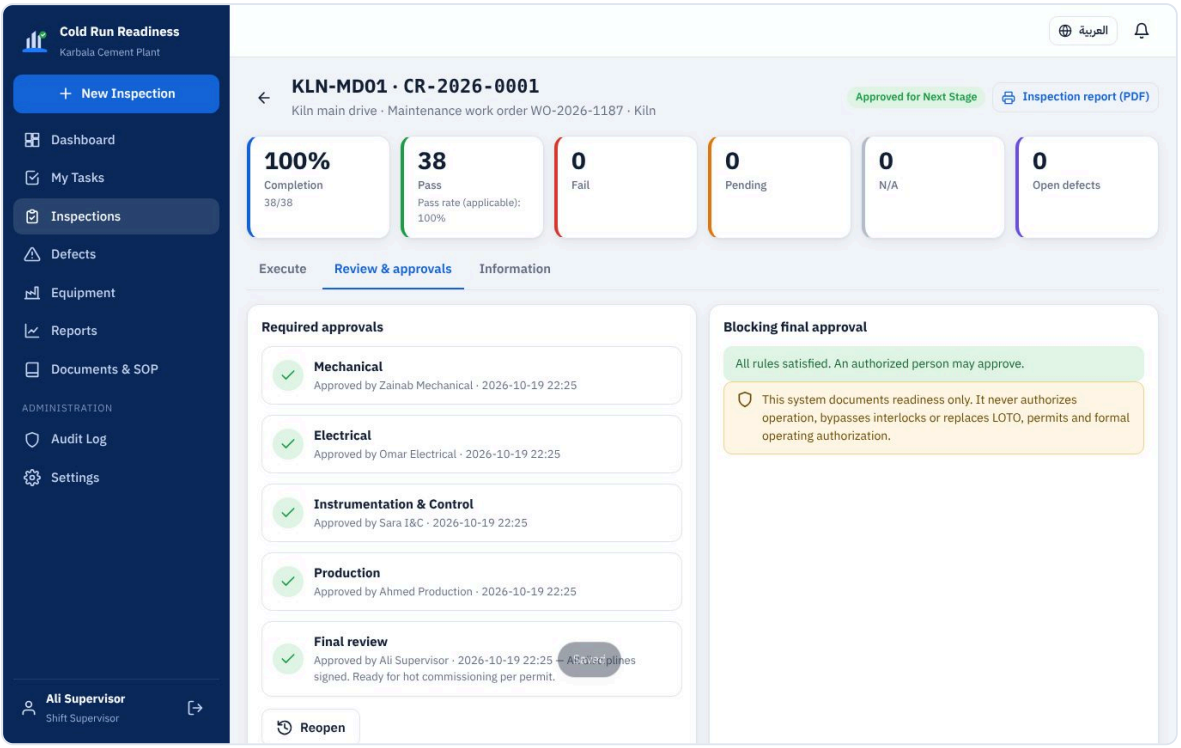


Awaiting verification — the person who repaired cannot verify their own repair.

6.6 Departmental approvals and final review

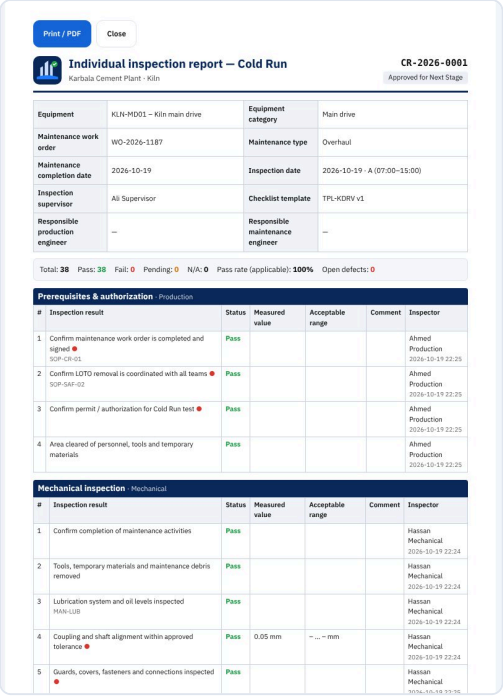


Review & approvals — four departmental sign-offs recorded with name and time; the final review becomes available when no blockers remain.

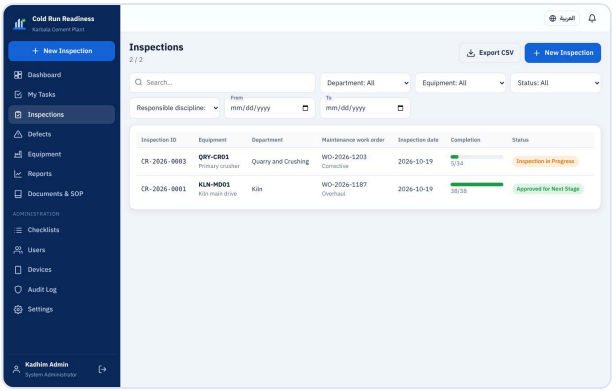


Approved for Next Stage — final approval with comment; authorized roles can reopen.

6.7 Inspection report (PDF)

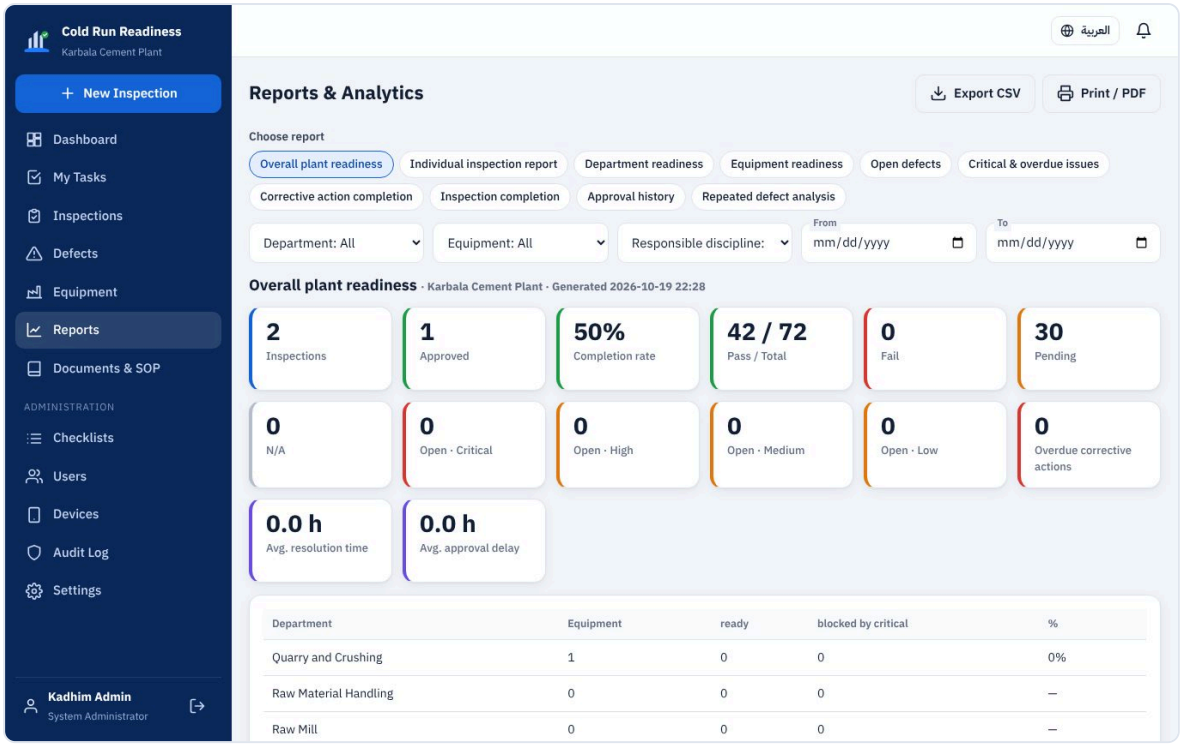

Print / PDF
Close
Individual inspection report — Cold Run
Karbala Cement Plant · Kiln
CR-2026-0001
Approved for Next Stage
Equipment
KLN-MD01 – Kiln main drive
Equipment category
Main drive
Maintenance work order
WO-2026-1187
Maintenance type
Overhaul
Maintenance completion date
2026-10-19
Inspection date
2026-10-19 · A (07:00–15:00)
Inspection supervisor
Ali Supervisor
Checklist template
TPL-KDRV v1
Responsible production engineer
–
Responsible maintenance engineer
–
Total: 38 Pass: 38 Fail: 0 Pending: 0 N/A: 0 Pass rate (applicable): 100% Open defects: 0
Prerequisites & authorization · Production
Inspection result Status Measured value Acceptable range Comment Inspector
1 Confirm maintenance work order is completed and signed SOP-CR-01 Pass Ahmed Production 2026-10-19 22:25
2 Confirm LOTO removal is coordinated with all teams SOP-SAF-02 Pass Ahmed Production 2026-10-19 22:25
3 Confirm permit / authorization for Cold Run test Pass Ahmed Production 2026-10-19 22:25
4 Area cleared of personnel, tools and temporary materials Pass Ahmed Production 2026-10-19 22:25
Mechanical inspection · Mechanical
Inspection result Status Measured value Acceptable range Comment Inspector
1 Confirm completion of maintenance activities Pass Hassan Mechanical 2026-10-19 22:24
2 Tools, temporary materials and maintenance debris removed Pass Hassan Mechanical 2026-10-19 22:24
3 Lubrication system and oil levels inspected MAN-LUB Pass Hassan Mechanical 2026-10-19 22:24
4 Coupling and shaft alignment within approved tolerance Pass 0.05 mm – … – mm Hassan Mechanical 2026-10-19 22:24
5 Guards, covers, fasteners and connections inspected Pass Hassan Mechanical

Individual inspection report (A4) — plant and equipment details, work order, personnel, results with measurements and limits, defects, photos, approvals with signature column, final status, document version and generation time. Saved as PDF from the print dialog on any device.

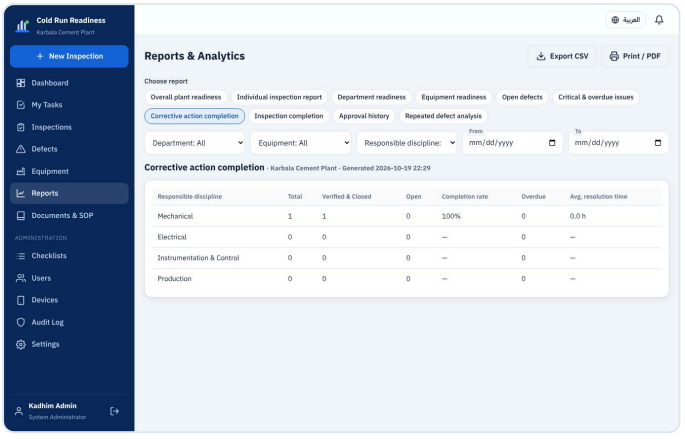

Cold Run Readiness
Karbala Cement Plant
New Inspection
Dashboard
My Tasks
Inspections
Defects
Equipment
Reports
Documents & SOP
ADMINISTRATION
Checklists
Users
Devices
Audit Log
Settings
Kadhim Admin
System Administrator
العربية
Inspections
2 / 2
Export CSV
New Inspection
Search…
Department: All
Equipment: All
Status: All
Responsible discipline:
From
mm/dd/yyyy
To
mm/dd/yyyy
Inspection ID Equipment Department Maintenance work order Inspection date Completion Status
CR-2026-0003 QRY-CR01 Primary crusher Quarry and Crushing WO-2026-1203 Corrective 2026-10-19 0/14 Inspection in Progress
CR-2026-0001 KLN-MD01 Kiln main drive Kiln WO-2026-1187 Overhaul 2026-10-19 38/38 Approved for Next Stage

Inspections history — searchable list with completion progress and status; CSV export; older records loaded on demand.

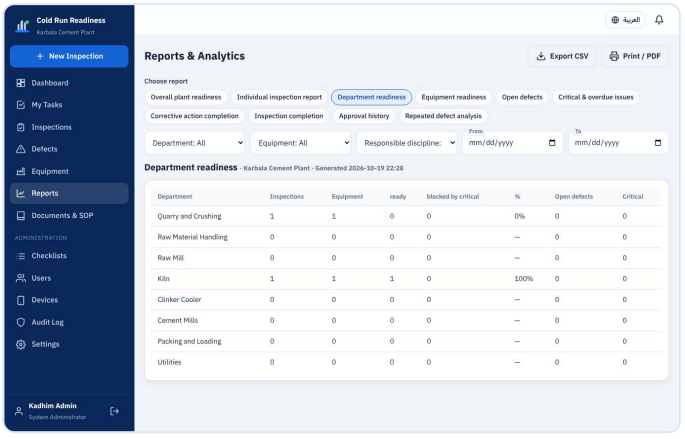
6.8 Reports and analytics



Overall plant readiness — totals, pass/fail/pending/N/A, open defects by severity, overdue actions, average resolution time and approval delay.



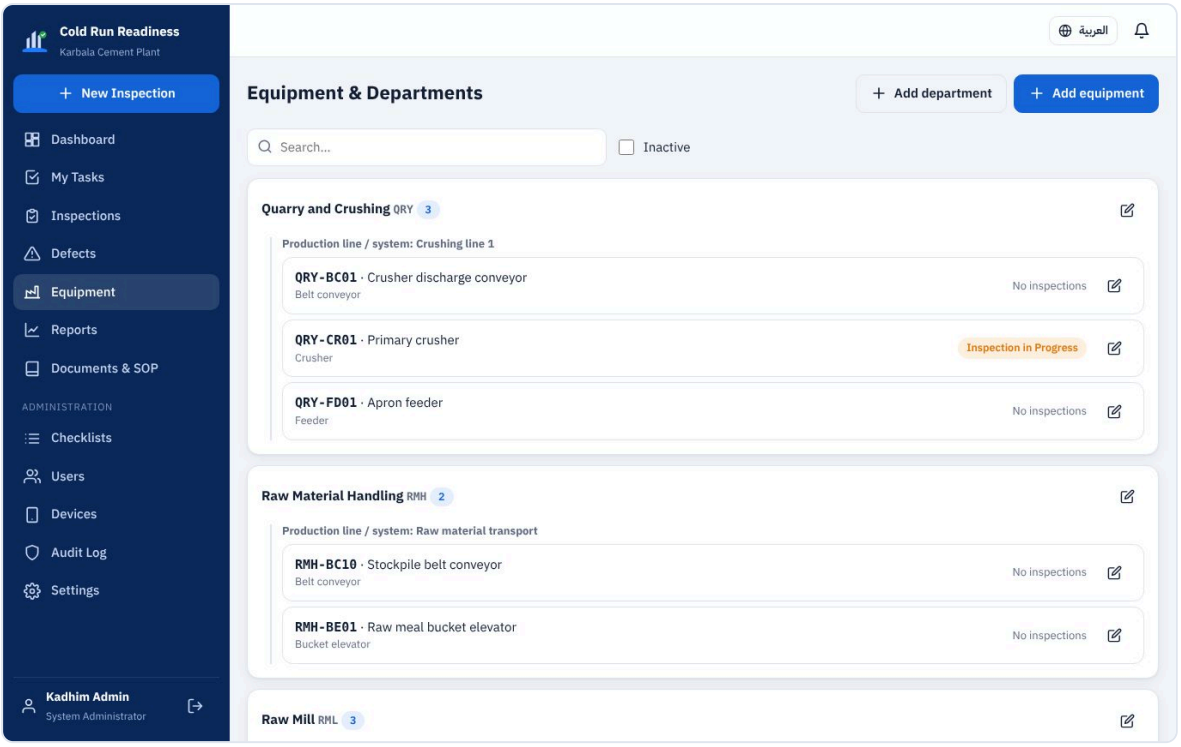
Corrective action completion by discipline.



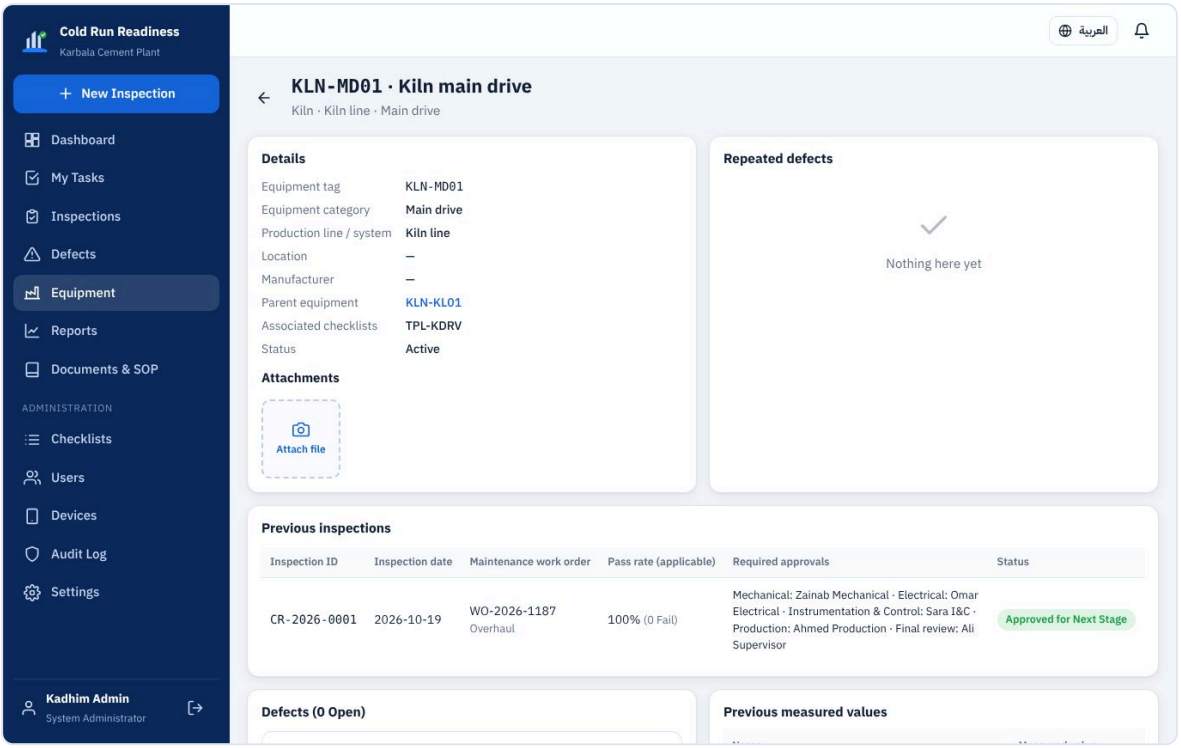
Department readiness report.

Available reports: overall plant readiness, individual inspection, department readiness, equipment readiness, open defects, critical & overdue issues, corrective action completion, inspection completion, approval history, repeated defect analysis.

6.9 Equipment hierarchy and history

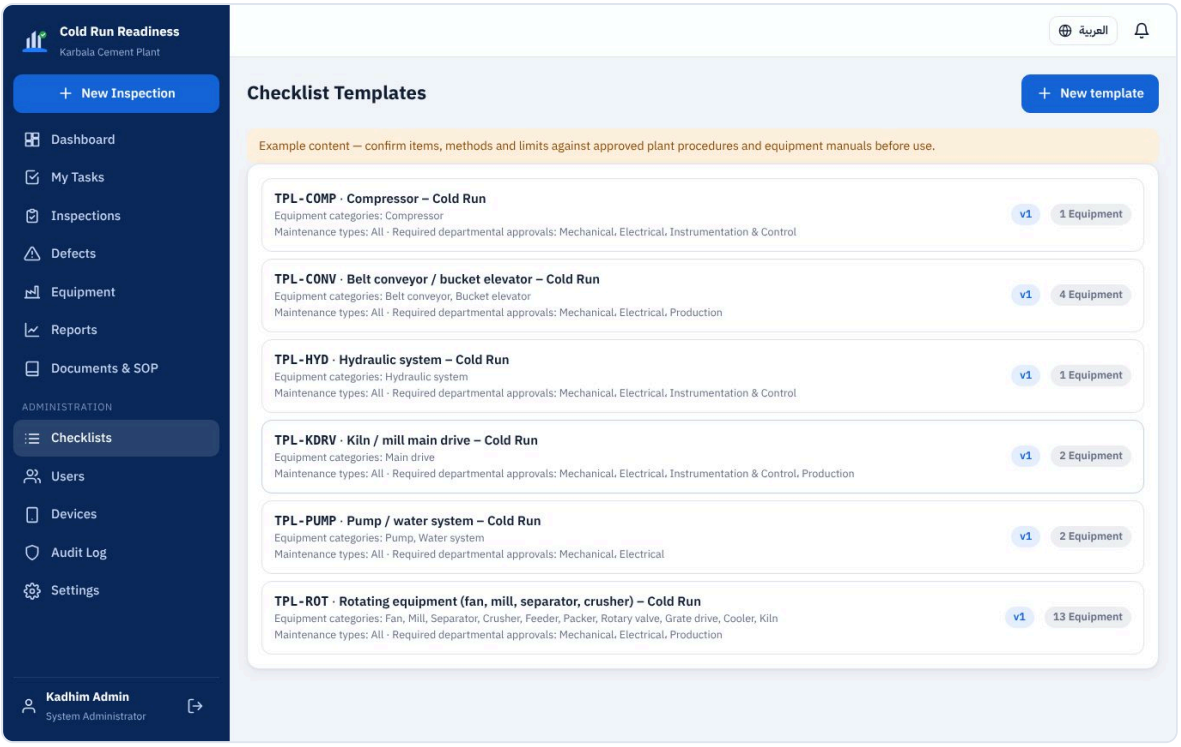


Equipment & departments — department → production line → equipment → sub-equipment, latest status and open defects; administrators add, edit, deactivate and re-parent without code changes.

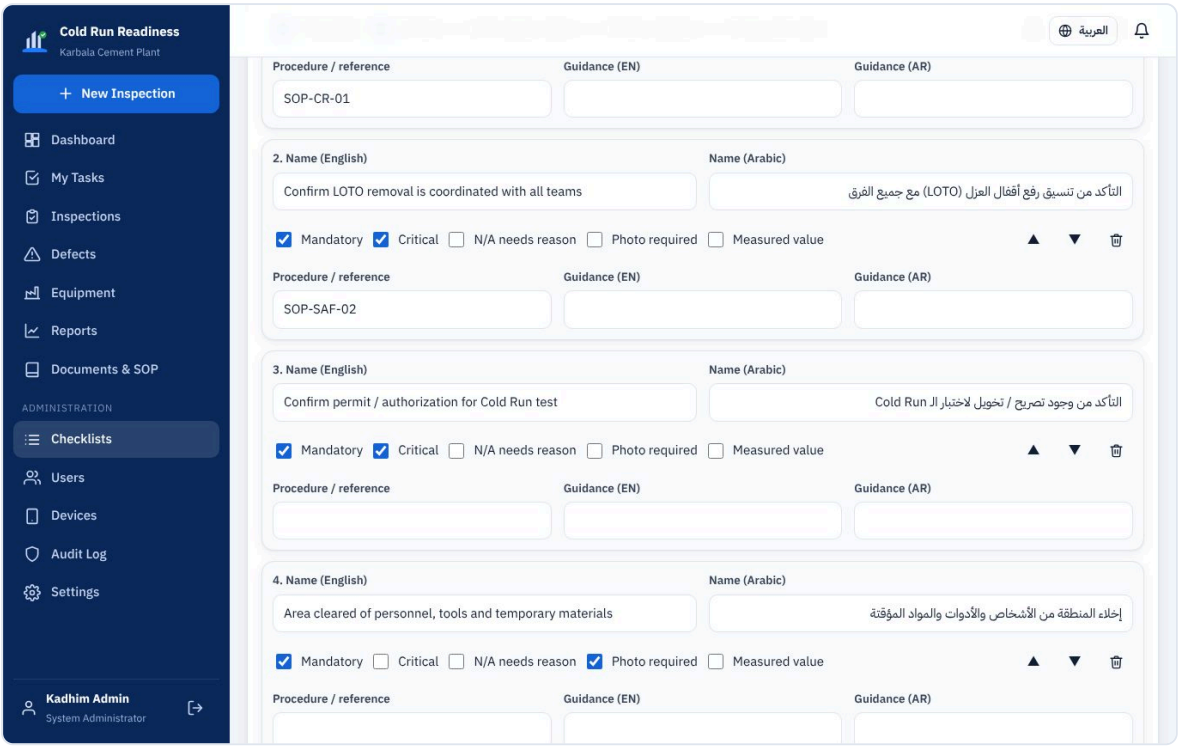


Equipment history — details, attachments, repeated defects, previous inspections with approvals, defects and previous measured values.

6.10 Checklist templates (versioned)

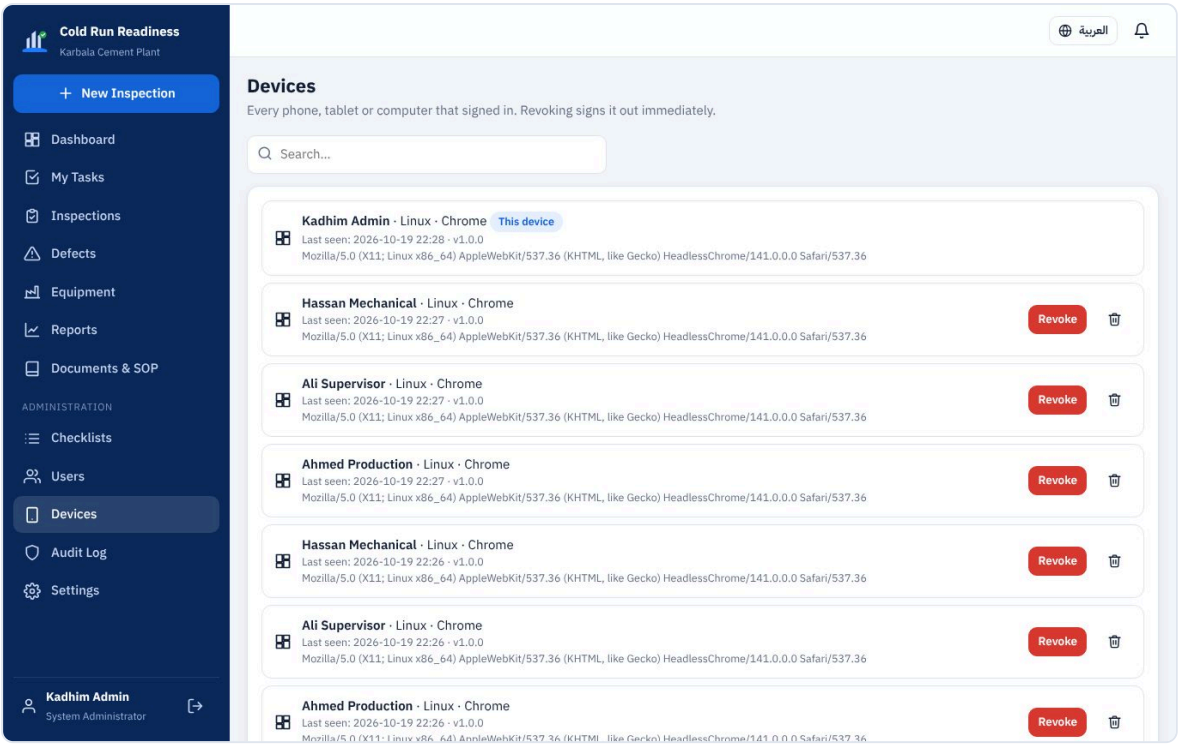


Templates — matched to equipment categories and maintenance types, with required departmental approvals and current version.

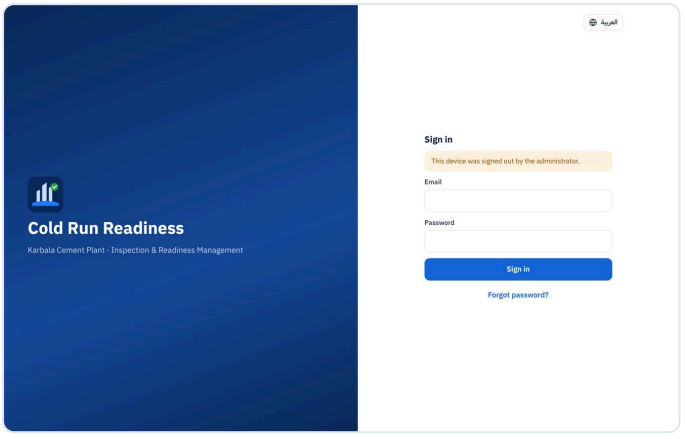


Template editor — bilingual sections and items, mandatory / critical / N/A-reason / photo / measurement settings, unit and limits, procedure reference, guidance. Publishing creates a new immutable version; running inspections keep the version they started with.

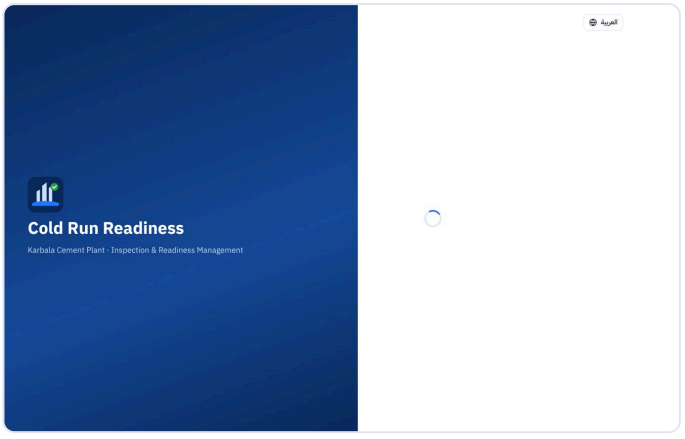
6.11 Devices, documents, audit and settings



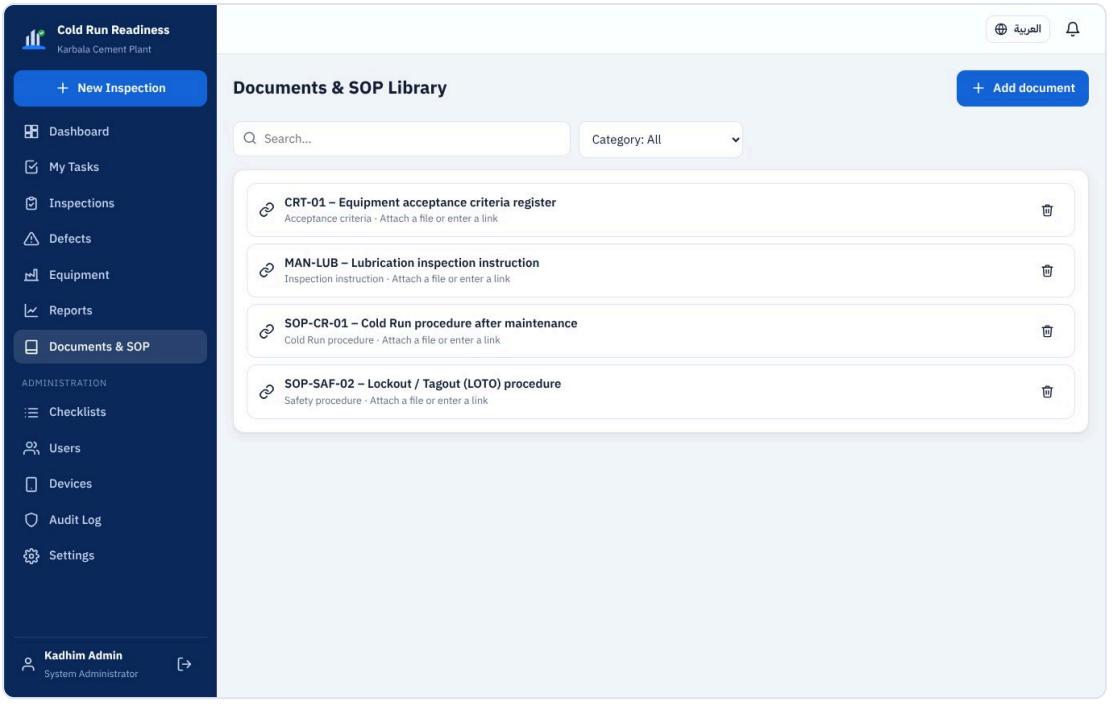
Devices — every phone/PC that signed in, with user, platform, app version and last seen. *Revoke* signs that device out immediately.



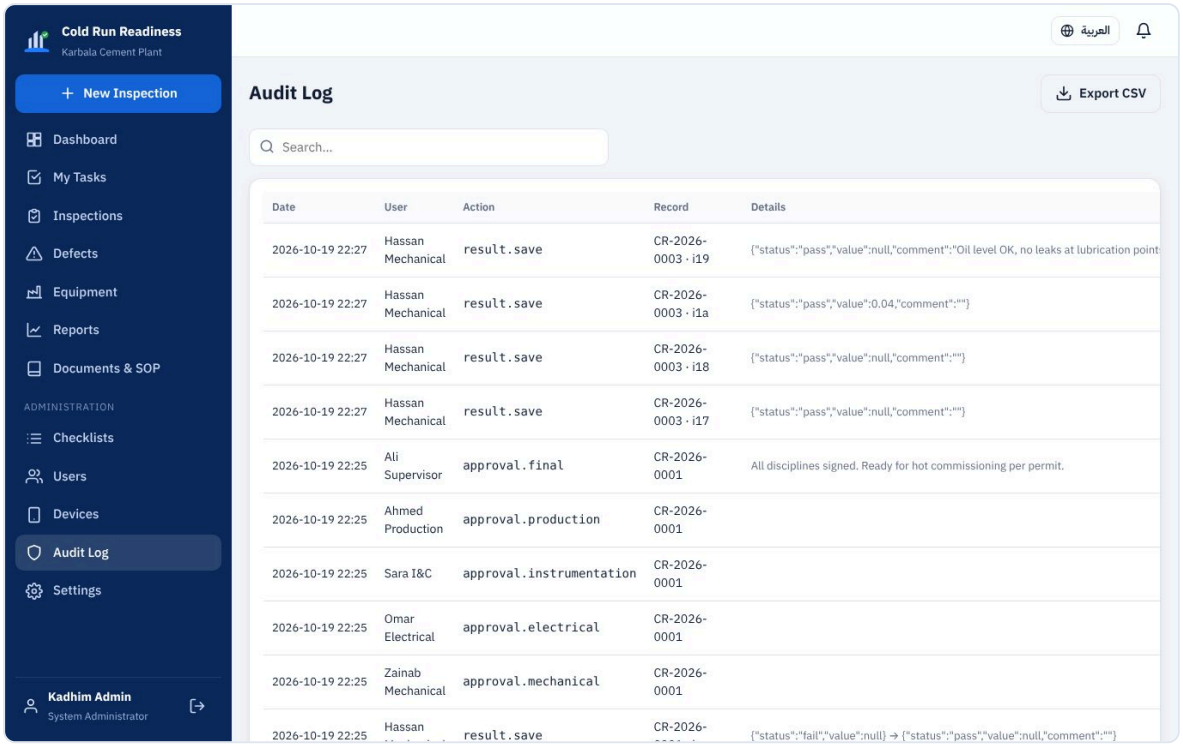
A revoked device is signed out with an explanation.



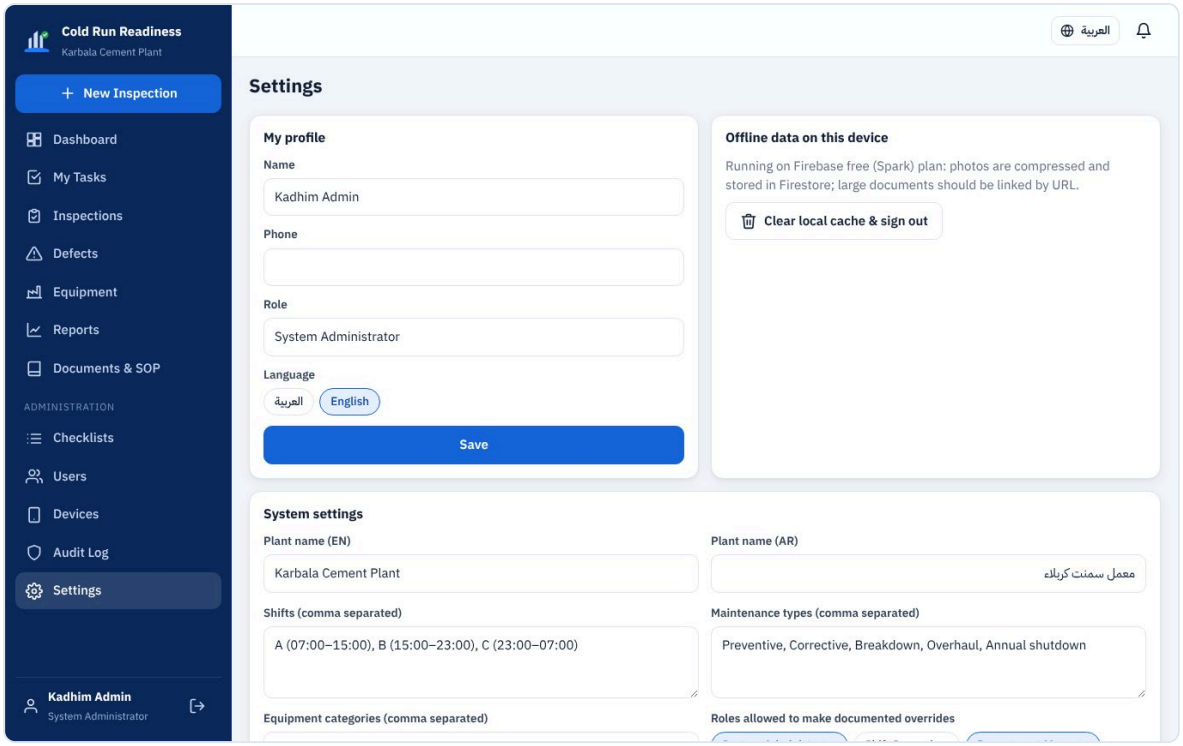
A deactivated user is blocked on all devices.



Documents & SOP library — procedures, manuals, instructions, acceptance criteria, safety procedures, forms and drawings; linked to department, equipment or template; visibility by role.



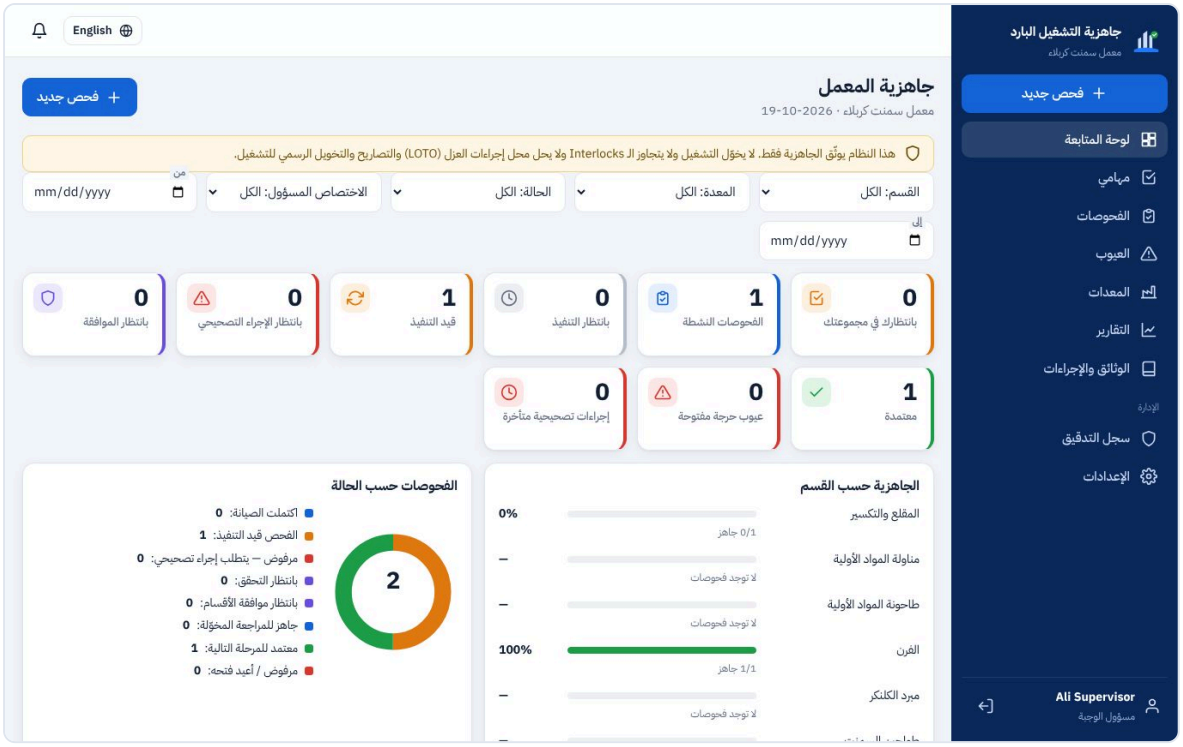
Audit log — append-only record of who created, edited, submitted, approved, rejected, reopened or closed each record, with before/after values; CSV export.



Settings — profile and language; plant name, shifts, maintenance types, equipment categories, override roles and target days per severity (administrator).

7. Arabic, mobile and offline use

Arabic uses a full right-to-left layout and English left-to-right; switching language keeps any data being entered. Standard industrial terms (Cold Run, DCS, Interlock, Coupling, LOTO) stay in English. On phones a bottom navigation bar and large touch targets are used.



Arabic dashboard (desktop, right-to-left).



Arabic inspection execution (desktop).



Arabic dashboard (phone).



My Tasks (phone).



Execution (phone).



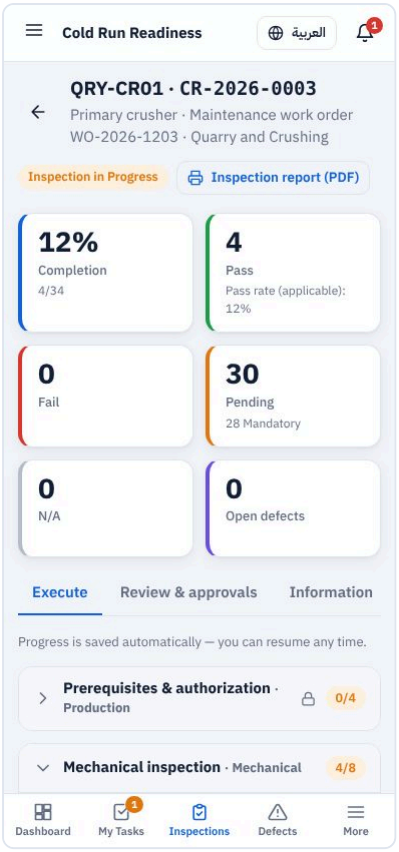
Fail with defect (phone).



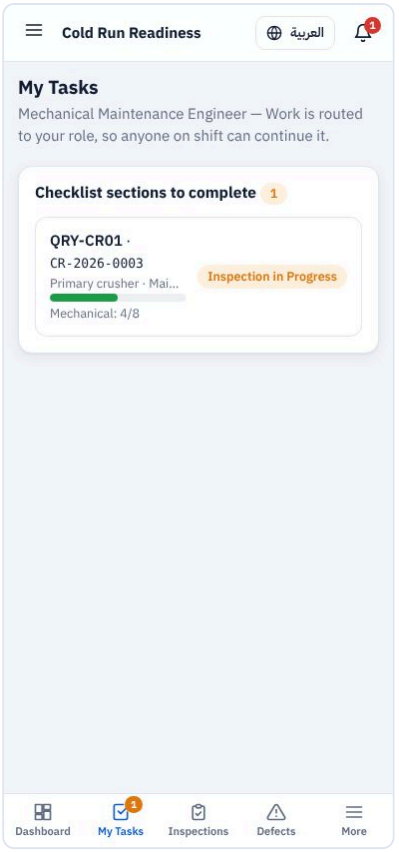
Offline: the indicator shows the result is saved on the device.



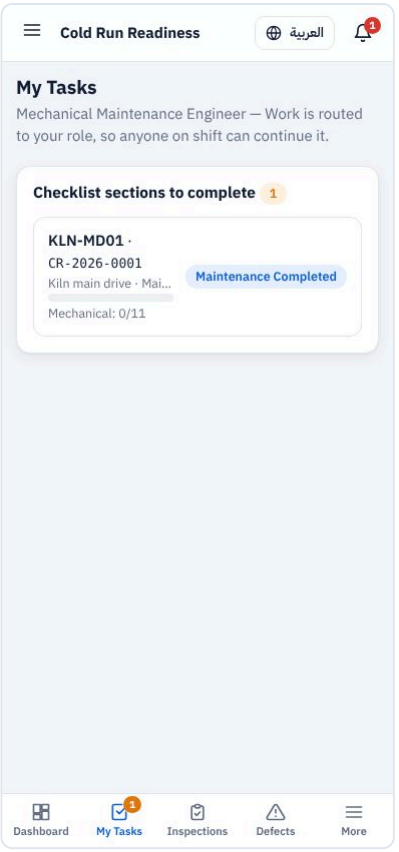
Side menu (phone).



English execution (phone).



English My Tasks (phone).



Mechanical engineer's pool.

Offline behaviour

- All data a user has seen is cached on the device; the app starts without network (installed PWA).
- Results, defects and approvals entered offline are queued and synchronised automatically when the connection returns (verified in testing).
- Conflicts are never silently overwritten: each item result has a revision number; an offline edit based on an older revision is rejected by the server, and online users get a "someone else updated this item" dialog to keep theirs or overwrite.
- The new-inspection form keeps an automatic draft on the device.

8. Security, approvals and audit trail

Because the free plan has no server code, every rule below is enforced by Firestore Security Rules on Google's servers — a modified app or direct API call cannot bypass them.

Rule	Enforcement
Only the first account can become administrator	Bootstrap allowed once (`config/setup`)
Users cannot change their own role; deactivated users lose all access	Profile fields restricted; every rule checks `active`
Only supervisors, managers and admins create inspections	Create rule by role
Engineers only record results for their own discipline	Result discipline must match role
No silent overwrite of results or inspections	Revision / document version must advance by one
Departmental approval only by that discipline (or manager/admin), in own name	Approval key and signer checked
Final approval only by leads, blocked by critical items unless documented override by configured roles	Status transition rule + settings
Defects: only leads close, only with passing verification, not by the repairer; repairs only recorded in own name	Defect transition rules
Audit log cannot be edited or deleted	Create-only, signed by the author with server time
Revoked devices cannot re-enable themselves	Only admins change `revoked`
Viewers are read-only; documents visible by role	Role checks on every write; document visibility list

9. Testing performed and results

12 / 12	4	9	0
security-rule test groups passed	end-to-end browser scenarios passed	user accounts / roles exercised	console errors in final runs

End-to-end scenarios (real app against local Firebase emulators, headless Chromium)

1. **Setup:** first administrator, example data, 8 user accounts created from the Users page.
2. **Full workflow:** supervisor creates a Kiln main drive inspection → mechanical engineer executes on a phone and fails a critical item (defect raised) → production, electrical and I&C complete their sections → second mechanical engineer takes the defect from the pool and repairs → mechanical engineer has no Verify button → a different supervisor verifies and closes → item re-inspected → four departmental approvals → final approval. Verified: mechanical engineer has no buttons on the electrical section.
3. **Arabic mobile & offline:** production engineer passes an item with the network off; offline indicator shown; synchronised after reconnecting.
4. **Administration:** reports, PDF generation, devices, audit, templates; revoking a device and deactivating a user signs both out immediately with the correct message.

Issues found and fixed during testing

- A Firestore SDK internal error (v11) under the offline cache — upgraded to Firebase JS SDK 12.19, error gone.
- A security rule using an unsupported function (`toList`) — rewritten; repair records restricted to the signer's own name.
- Creations now wait for server confirmation when online (no record left only on a device that is closed immediately).
- Sections with failed items open automatically; stale failure comment cleared on re-inspection; date filter labels; search icon overlap; draft-restored message only for real drafts.

Automation: `.github/workflows/platform-ci.yml` runs type-check, production build and the security-rule tests on every change to `platform/`.

10. Deployment to Firebase Hosting

The workflow `.github/workflows/firebase-deploy.yml` builds the app and deploys Hosting, Firestore rules and indexes on every merge to `main` (or manually from GitHub → Actions). One-time setup:

1. **Create the Firebase project** (console.firebase.google.com, Spark plan): enable *Authentication → Email/Password*; create a *Firestore database* in production mode; register a *Web app* and note `apiKey`, `projectId`, `appId`, `messagingSenderId`.
2. **Service account key**: *Project settings → Service accounts → Generate new private key*; in Google Cloud IAM give that account the **Firebase Admin** role.
3. **GitHub secrets** (*repository → Settings → Secrets and variables → Actions*):

Secret	Value
`FIREBASE_SERVICE_ACCOUNT`	Entire content of the JSON key file
`FIREBASE_PROJECT_ID`	projectId
`FIREBASE_API_KEY`	apiKey
`FIREBASE_APP_ID`	appId
`FIREBASE_MESSAGING_SENDER_ID`	messagingSenderId

4. **Publish**: merge the branch into `main` (or run the workflow). The run summary shows `https://<project-id>.web.app`.
5. **First use**: open the link, create the administrator, load and review the example data, add users. On phones choose *Add to Home Screen / Install app*.

Alternative (from a computer): in `platform/` create `.env.local` from `.env.example`, set the project in `.firebaserc`, then run `npm install`, `npx firebase login`, `npm run deploy`.

11. Limitations and recommended next steps

Current limitations (v1, free plan)

- Not yet published: requires your Firebase project and the GitHub secrets above.
- Example checklists, methods and limits must be replaced or confirmed by plant engineering and safety personnel.
- PDFs are produced through the browser's print dialog ("Save as PDF").
- Notifications are in-app only; push/e-mail need Cloud Functions (Blaze plan).
- Files are limited to ≈650 KB each; large manuals should be linked by URL.
- Deactivated users remain in Firebase Authentication (blocked by rules) because deleting accounts needs the Admin SDK.
- Free-plan daily quotas (50k reads / 20k writes / 1 GiB storage) suit a single plant; heavy photo use may require Blaze later.

Recommended next steps

1. Create the Firebase project and secrets; merge to `main` to publish.
2. Plant engineering to review departments, equipment register and checklist templates (items, critical flags, units and limits, required approvals).
3. Create real user accounts and assign department scopes; pilot on one department (e.g. Kiln) for two weeks.
4. Optional upgrade to Blaze for Cloud Storage photos, push/e-mail notifications and server-generated PDFs.

> This application is a documentation and workflow tool. It is not a replacement for physical safety systems, lockout/tagout, permits or formal plant operating authorization, and it never energizes equipment or bypasses interlocks.